

Chapter 4, Section 2

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June 1, 2026

1. Use (2.5.3) to show that $\mathbf{P}^n$ is simply connected.

Proof. Let $f : X \rightarrow \mathbf{P}^n$ be an étale covering. We may assume X is connected. It suffices to show f is one-to-one. Let y be any closed point in $\mathbf{P}^n$, and let $L = \mathbf{P}^1$ be any line in $\mathbf{P}^n$ containing y . The restriction $f : f^{-1}(L) \rightarrow L$, obtained by taking the base change $L \hookrightarrow \mathbf{P}^n$, is an étale covering, so $f^{-1}(L)$ is a disjoint union of rational curves (2.5.3). $\square$

2. *Classification of Curves of Genus 2.* Fix an algebraically closed field k of characteristic $\neq 2$.

- (a) If X is a curve of genus 2 over k , the canonical linear system $|K|$ determines a finite morphism $f : X \rightarrow \mathbf{P}^1$ of degree 2 (Ex. 1.7). Show that it is ramified at exactly 6 points, with ramification index 2 at each one. Note that f is uniquely determined, up to an automorphism of $\mathbf{P}^1$, so X determines an (unordered) set of 6 points of $\mathbf{P}^1$, up to an automorphism of $\mathbf{P}^1$.
- (b) Conversely, given six distinct elements $\alpha_1, \dots, \alpha_6 \in k$, let K be the extension of $k(x)$ determined by the equation $z^2 = (x - \alpha_1) \cdots (x - \alpha_6)$. Let $f : X \rightarrow \mathbf{P}^1$ be the corresponding morphism of curves. Show that $g(X) = 2$, the map f is the same as the one determined by the canonical linear system, and f is ramified over the six points $x = \alpha_i$ of $\mathbf{P}^1$, and nowhere else.
- (c) Using (I, Ex. 6.6), show that if P_1, P_2, P_3 are three distinct points of $\mathbf{P}^1$, then there exists a unique $\varphi \in \text{Aut } \mathbf{P}^1$ such that $\varphi(P_1) = 0$, $\varphi(P_2) = 1$, $\varphi(P_3) = \infty$. Thus in (a), if we order the six points of $\mathbf{P}^1$, and then normalize by sending the first three to $0, 1, \infty$, respectively, we may assume that X is ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$, where $\beta_1, \beta_2, \beta_3$ are three distinct elements of k , $\neq 0, 1$.
- (d) Let Σ_6 be the symmetric group on 6 letters. Define an action of Σ_6 on sets of three distinct elements $\beta_1, \beta_2, \beta_3$ of k , $\neq 0, 1$, as follows: reorder the set $0, 1, \infty, \beta_1, \beta_2, \beta_3$ according to a given element $\sigma \in \Sigma_6$, then renormalize as in (c) so that the three become $0, 1, \infty$ again. Then the last three are the new $\beta'_1, \beta'_2, \beta'_3$.
- (e) Summing up, conclude that there is a one-to-one correspondence between the set of isomorphism classes of curves of genus 2 over k , and triples of distinct elements $\beta_1, \beta_2, \beta_3$ of k , $\neq 0, 1$, modulo the action of Σ_6 described in (d). In particular, there are many non-isomorphic curves of genus 2. We say that curves of genus 2 depend on three parameters, since they correspond to the points of an open subset of $\mathbf{A}_k^3$ modulo a finite group.

Proof.

- (a) By Hurwitz's theorem, the degree of the ramification divisor is computed to be

$$\deg R = 2g(X) - 2 - n(2g(\mathbf{P}^1) - 2) = 6.$$

The ramification index at a point cannot exceed the degree of the morphism, so f must be ramified at exactly 6 distinct points each having ramification index 2.

- (b) $B = k[x, z]/(z^2 - g(x))$ and $g(x) = (x - \alpha_1) \cdots (x - \alpha_6)$, and f is the projection morphism onto the x -axis. By (II, 8.3A), we have

$$\Omega_{B/k} \cong (Bdx \oplus Bdz)/(g'(x)dx + 2zdz).$$

Let $P = (x', z')$ be a closed point on X . If $z' \neq 0$, then z is invertible in $\mathcal{O}_{P,X}$, so that

$$dz = -\frac{g'(x)}{2z}dx,$$

which means dx generates $(\Omega_{B/k})_P$. If $z' = 0$, then $x' = \alpha_i$ for some i , and $g'(x)$ is invertible in $\mathcal{O}_{P,X}$, so that

$$dx = -\frac{2z}{g'(x)}dz,$$

which means dz generates $(\Omega_{B/k})_P$. Since dx is a local parameter of $\mathbf{P}^1$, we see that P is ramified if and only if $z' = 0$, with $dx/dz = -2z/g'(z)$. Moreover, $v_P(dx/dz) = 1$, so the ramification index is 2 (2.2), and the degree of the ramification divisor is 6. By Hurwitz's theorem, $g(X) = 2$.

- (c) Let x be an affine coordinate on $\mathbf{P}^1$ such that $P_i \neq \infty$ and identify P_i with some $\alpha_i \in k$. The following automorphism will do

$$\varphi(x) = \frac{\alpha_2 - \alpha_3}{\alpha_2 - \alpha_1} \frac{x - \alpha_1}{x - \alpha_3}.$$

- (e) We want to show two curves of genus 2 are isomorphic if and only if they correspond to the same triples of distinct elements in k , $\neq 0, 1$, described in (d).

□

5. Automorphisms of a Curve of Genus ≥ 2 . Prove the theorem of Hurwitz [1] that a curve X of genus $g \geq 2$ over a field of characteristic 0 has at most $84(g - 1)$ automorphisms. We will see later (Ex. 5.2) or (V, Ex. 1.11) that the group $G = \text{Aut } X$ is finite. So let G have order n . Then G acts on the function field $K(X)$. Let L be the fixed field. Then the field extension $L \subseteq K(X)$ corresponds to a finite morphism of curves $f : X \rightarrow Y$ degree n .

- (a) If $P \in X$ is a ramification point, and $e_P = r$, show that $f^{-1}f(P)$ consists of exactly n/r points, each having ramification index r . Let $P_1, \dots, P_s$ be a maximal set of ramification points of X lying over distinct points of Y , and let $e_{P_i} = r_i$. Then show that Hurwitz's theorem implies that

$$(2g - 2)/n = 2g(Y) - 2 + \sum_{i=1}^s (1 - 1/r_i).$$

- (b) Since $g \geq 2$, the left hand side of the equation is > 0 . Show that if $g(Y) \geq 0$, $s \geq 0$, $r_i \geq 2$, $i = 1, \dots, s$ are integers such that

$$2g(Y) - 2 + \sum_{i=1}^s (1 - 1/r_i) > 0,$$

then the minimum value of this expression is $1/42$. Conclude that $n \geq 84(g - 1)$.

Proof.

- (a)
(b) The following give the minimum value

$$g(Y) = 0, \quad s = 3, \quad r_1 = \frac{1}{2}, \quad r_2 = \frac{1}{3}, \quad r_3 = \frac{1}{7}.$$

We conclude that

$$n \leq 42(2g - 2) = 84(g - 1).$$

□

6. *f_* for Divisors.* Let $f : X \rightarrow Y$ be a finite morphism of curves of degree n . We define a homomorphism $f_* : \text{Div } X \rightarrow \text{Div } Y$ by $f_*(\sum n_i P_i) = \sum n_i f(P_i)$ for any divisor $D = \sum n_i P_i$ on X .

- (a) For any locally free sheaf $\mathcal{E}$ on Y , of rank r , we define $\det \mathcal{E} = \wedge^r \mathcal{E} \in \text{Pic } Y$ (II, Ex. 6.11). In particular, for any invertible sheaf $\mathcal{M}$ on X , $f_* \mathcal{M}$ is locally free of rank n on Y , so we can consider $\det f_* \mathcal{M} \in \text{Pic } Y$. Show that for any divisor D on X ,

$$\det(f_* \mathcal{O}_X(D)) \cong (\det f_* \mathcal{O}_X) \otimes \mathcal{O}_Y(f_* D).$$

Note in particular that $\det(f_* \mathcal{O}_X(D)) \neq \mathcal{O}_Y(f_* D)$ in general!

- (b) Conclude that $f_* D$ depends only on the linear equivalence class of D , so there is an induced homomorphism $f_* : \text{Pic } X \rightarrow \text{Pic } Y$. Show that $f_* \circ f^* : \text{Pic } Y \rightarrow \text{Pic } Y$ is just multiplication by n .
(c) Use duality for a finite flat morphisms (III, Ex. 6.10) and (III, Ex. 7.2) to show that

$$\det f_* \Omega_X \cong (\det f_* \mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

- (d) Now assume that f is separable, so we have the ramification divisor R . We define the *branch divisor* B to be the divisor $f_* R$ on Y . Show that

$$(\det f_* \mathcal{O}_X)^2 \cong \mathcal{O}_Y(-B).$$

Proof.

- (a) Consider the case when D is an effective divisor. The pushforward f_* is exact for finite morphisms, so applying f_* to the exact sequence

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X(D) \longrightarrow \mathcal{O}_D \longrightarrow 0$$

gives us another exact sequence

$$0 \longrightarrow f_*\mathcal{O}_X \longrightarrow f_*\mathcal{O}_X(D) \longrightarrow f_*\mathcal{O}_D \longrightarrow 0.$$

Taking determinants, we have

$$\det(f_*\mathcal{O}_X(D)) \cong (\det f_*\mathcal{O}_X) \otimes \det(f_*\mathcal{O}_D)$$

- (b) It suffices to show $f_*f^*Q = nQ$ for every point $Q \in Y$. By definition, $f_*f^*Q = n'Q$ for some $n' \geq 0$. On the otherhand, $\deg f_*D = \deg D$, and $\deg f^*D = n \deg D$ (II, 6.9), so $\deg f_*f^*D = n \deg D$. Hence, $n' = n$.
- (c)
- (d) Let K_X and K_Y be canonical divisors of X and Y , respectively. They satisfy the following relation (2.3)

$$f^*K_Y - K_X \sim -R.$$

Applying f_* gives

$$nK_Y - f_*K_X \sim -B,$$

where we used the fact that $f_*f^*K_Y = nK_Y$ (b). On the otherhand, the results of (a) applied to $D = K_X$ and (c) give

$$\mathcal{O}_Y(nK_Y) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X), \quad \mathcal{O}_Y(-f_*K_X) \cong (\det f_*\Omega_X) \otimes (\det f_*\Omega_X)^{-1},$$

so combining everything, we get

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(nK_Y - f_*K_X) \cong \mathcal{O}_Y(-B).$$

□

7. Étale Covers of Degree 2. Let Y be a curve over a field k of characteristic $\neq 2$. We show there is a one-to-one correspondence between finite étale morphisms $f : X \rightarrow Y$ of degree 2, and 2-torsion elements of $\text{Pic } Y$, i.e., invertible sheaves $\mathcal{L}$ on Y with $\mathcal{L}^2 \cong \mathcal{O}_Y$.

- (a) Given an étale morphism $f : X \rightarrow Y$ of degree 2, there is a natural map $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. Let $\mathcal{L}$ be the cokernel. Then $\mathcal{L}$ is an invertible sheaf on Y , $\mathcal{L} \cong \det f_*\mathcal{O}_X$, and so $\mathcal{L}^2 \cong \mathcal{O}_Y$ by (Ex. 2.6). Thus an étale cover of degree 2 determines a 2-torsion element in $\text{Pic } Y$.
- (b) Conversely, given a 2-torsion element $\mathcal{L}$ in $\text{Pic } Y$, define an $\mathcal{O}_Y$ -algebra structure on $\mathcal{O}_Y \oplus \mathcal{L}$ by $\langle a, b \rangle \cdot \langle a', b' \rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b \rangle$, where φ is an isomorphism of $\mathcal{L} \otimes \mathcal{L} \rightarrow \mathcal{O}_Y$. Then take $X = \mathbf{Spec}(\mathcal{O}_Y \oplus \mathcal{L})$. Show that X is an étale cover of Y .

(c) Show that these two processes are inverse to each other.

Proof.

(b) For each point $y \in Y$, we have $X_y \cong \operatorname{Spec}(\mathcal{O}_y^{\oplus 2})$ with $\mathcal{O}_y$ -algebra structure given by

$$\langle a, b \rangle \cdot \langle a', b' \rangle = \langle aa' + bb', ab' + a'b \rangle.$$

In particular, the natural morphism $f : X \rightarrow Y$ is flat because $f_*\mathcal{O}_X$ is locally free of rank 2, and the structure morphism $\mathcal{O}_y \rightarrow \mathcal{O}_y^{\oplus 2}$ is given by $1 \mapsto \langle 1, 0 \rangle$.

(c)

□